

PRACTICAL (OSSP)

SESSION – 3

1. Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

```
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_2$ cd ~/OSSP/PRACTICAL/Session_3
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ pwd
/home/koumudi/OSSP/PRACTICAL/Session_3
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ nano process_states.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ gcc process_states.c -o process_states
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ ./process_states
Parent process started.
Parent PID = 1486
Parent PPID = 549

--- PARENT PROCESS ---
Parent PID = 1486
Parent waiting for child...

--- CHILD PROCESS ---
Child PID = 1487
Child PPID = 1486
Child process is running.
Child process terminating.
Parent process continuing after child termination.
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ |
```

2. Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

```
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ nano state_demo.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ gcc state_demo.c -o state_demo
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ ./state_demo &
[1] 1587
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ Process started. PID = 1587
Process is running...
ps -o pid,ppid,stat,cmd -p 1587
  PID    PPID  STAT  CMD
  1587    549   S     ./state_demo
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ |
```

```
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ Process started. PID = 1614
Process is running...
cat /proc/1614/status | grep -E "Name|State|cat /proc/1614/status | grep -E "Name|State|Pid|PPid"
Name:    state_demo
State:   S (sleeping)
Pid:     1614
PPid:    549
TracerPid: 0
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ Process terminating...
```

```
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ Process terminating...
top
top - 17:17:18 up 1:18, 1 user, load average: 0.08, 0.04, 0.00
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7784.6 total, 7063.4 free, 516.1 used, 360.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7268.5 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21876	13156	9668	S	0.0	0.2	0:02.02	systemd
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.04	init-systemd(Ub
6	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
45	root	19	-1	66828	19628	18512	S	0.0	0.2	0:00.72	systemd-journal
92	root	20	0	25020	6404	5180	S	0.0	0.1	0:01.54	systemd-udev
110	systemd+	20	0	21344	13236	11112	S	0.0	0.2	0:00.45	systemd-resolve
111	systemd+	20	0	91036	7964	6996	S	0.0	0.1	0:00.35	systemd-timesyn
171	root	20	0	4244	2688	2432	S	0.0	0.0	0:00.05	cron
172	message+	20	0	9648	5340	4608	S	0.0	0.1	0:00.29	dbus-daemon
185	root	20	0	17980	8840	7800	S	0.0	0.1	0:00.36	systemd-logind
208	syslog	20	0	222516	5956	4576	S	0.0	0.1	0:00.32	rsyslogd
215	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.01	agetty
222	root	20	0	107036	23016	13560	S	0.0	0.3	0:00.51	unattended-upgr
343	root	20	0	6704	4652	3872	S	0.0	0.1	0:00.02	login
391	koumudi	20	0	20320	11572	9452	S	0.0	0.1	0:00.24	systemd
392	koumudi	20	0	21160	3592	1868	S	0.0	0.0	0:00.00	(sd-pam)
411	koumudi	20	0	6080	5340	3672	S	0.0	0.1	0:00.09	bash
547	root	20	0	3184	1124	980	S	0.0	0.0	0:00.00	SessionLeader
548	root	20	0	3200	1260	1108	S	0.0	0.0	0:00.11	Relay(549)
549	koumudi	20	0	6596	5948	3700	S	0.0	0.1	0:00.16	bash
1256	polkitd	20	0	308172	8152	7256	S	0.0	0.1	0:00.11	polkitd
1631	koumudi	20	0	9300	5680	3496	R	0.0	0.1	0:00.01	top

```
[1]+ Done ./state_demo
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ ps -p 1614
PID TTY TIME CMD
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_3$ |
```